

Path 2: Cyber Security 101 > Module 4: Command Line > Room 1: Windows Command Line

<https://tryhackme.com/room/windowscommandline>

Windows Command Line:

Learn the essential Windows commands.

>> Task 1: Introduction

Everyone prefers a graphical user interface (GUI) until they master a command-line interface (CLI). There are many reasons for that. One reason is that GUIs are usually intuitive. If someone offers you a GUI interface you are unfamiliar with, you can quickly poke around and discover a non-trivial part. Compare this with dealing with a CLI, i.e., a prompt.

CLI interfaces usually have a learning curve; however, as you master the command line, you will find it faster and more efficient. Consider this trivial example: How many clicks do you need to find your IP address using the graphical desktop? Using the command-line interface, you don't even need to raise your hands off the keyboard. Let's say you want to recheck your IP address. You need to issue the same command instead of moving the mouse pointer to every corner of your screen.

There are many other advantages to using a CLI besides speed and efficiency. We will mention a few:

- **Lower resource usage:** CLIs require fewer system resources than graphics-intensive GUIs. In other words, you can run your CLI system on older hardware or systems with limited memory. If you are using cloud computing, your system will require lower resources, which in turn will lower your bill.
- **Automation:** While you can automate GUI tasks, creating a batch file or script with the commands you need to repeat is much easier.
- **Remote management:** CLI makes it very convenient to use SSH to manage a remote system such as a server, router, or an IoT device. This approach works well on slow network speeds and systems with limited resources.

♦ Learning Objectives:

The purpose of this room is to teach you how to use MS Windows Command Prompt cmd.exe, the default command-line interpreter in the Windows environment. We will learn how to use the command line to:

- Display basic system information
- Check and troubleshoot network configuration
- Manage files and folders
- Check running processes

♦ Room Prerequisites:

Before starting this room, you should have finished the "Windows and AD Fundamentals module"

(<https://tryhackme.com/module/windows-and-active-directory-fundamentals>).

Q1) What is the default command line interpreter in the Windows environment?

Answer: cmd.exe

>> Task 2: Basic System Information

Before issuing commands, we should note that we can only issue the commands within the Windows Path. You can issue the command **set** to check your path from the command line. The terminal output shows the path where MS Windows will execute commands, as indicated by the line starting with **Path=**.

Let's use the **ver** command to determine the operating system (OS) version.

Enough warming up. Let's discover more in-depth information about the system. We can run the **systeminfo** command to list various information about the system such as OS information, system details, processor and memory. The terminal below shows a snippet of the displayed output.

Before moving on, it is good to mention a couple of tricks.

First, you can pipe it through **more** if the output is too long. Then, you can view it page after page by pressing the space bar button. To demonstrate this, try running **driverquery** and compare it with running **driverquery | more**. In the latter, you can display the output page by page and you can exit it using **CTRL + C**.

help - Provides help information for a specific command.

cls - Clears the Command Prompt screen.

Q1) What is the OS version of the Windows VM?

Answer: 10.0.20348.2655

Q2) What is the hostname of the Windows VM?

Answer: WINSRV2022-CORE

>> Task 3: Network Troubleshooting

Most of us are used to looking up MS Windows network configuration from the GUI interface. The command-line interface provides many networking-related commands to look up your current configuration, check ongoing connections, and troubleshoot networking issues.

- ◆ Network Configuration:

You can check your network information using `ipconfig`. The terminal output below shows our IP address, subnet mask, and default gateway.

You can also use `ipconfig /all` for more information about your network configuration. As shown in the terminal below, we can view our DNS servers and confirm that DHCP is enabled.

- ◆ Network Troubleshooting:

One common troubleshooting task is checking if the server can access a particular server on the Internet. The command syntax is `ping target_name`. Inspired by ping-pong, we send a specific ICMP packet and listen for a response. If a response is received, we know that we can reach the target and that the target can reach us.

Another valuable tool for troubleshooting is `tracert`, which stands for trace route. The command `tracert target_name` traces the network route traversed to reach the target. Without getting into more details, it expects the routers on the path to notify us if they drop a packet because its time-to-live (TTL) has reached zero.

- ◆ More Networking Commands

- One networking command worth knowing is "`nslookup`". It looks up a host or domain and returns its IP address. The syntax `nslookup example.com` will look up **example.com** using the default name server; however, `nslookup example.com 1.1.1.1` will use the name server **one.one.one.one**.

- The final networking command we will cover in this room is `netstat`. This command displays current network connections and listening ports. A basic `netstat` command with no arguments will show you established connections.

If you are curious about the other options, you can run `netstat -h`, where `-h` displays the help page. We opted for the following options

:

- `-a` displays all established connections and listening ports
- `-b` shows the program associated with each listening port and established connection
- `-o` reveals the process ID (PID) associated with the connection
- `-n` uses a numerical form for addresses and port numbers

We combine these four options and execute the `netstat -abon` command. The result is quite long, but we display the first few lines in the terminal below. It is clear now that the executable **sshd.exe** is responsible for listening for incoming connections on port 22. We can also see the process ID (PID) associated with each connection.

Q1) Which command can we use to look up the server's physical address (MAC address)?

Answer: `ipconfig /all`

Q2) What is the name of the service listening on port 135?

Answer: RpcSs

Q3) What is the name of the service listening on port 3389?

Answer: TermService

>> Task 4: File and Disk Management

♦ Working With Directories:

You can use **cd** without parameters to display the current drive and directory. It is the equivalent of asking the system, where am I?

You can view the child directories using **dir**.

Note that you can use the following options with **dir**:

- **dir /a** - Displays hidden and system files as well.
- **dir /s** - Displays files in the current directory and all subdirectories.

You can type **tree** to visually represent the child directories and subdirectories.

You can change to any directory by using the command **cd target_directory**; this is equivalent to double-clicking the **target_directory** on your desktop. Furthermore, you can use **cd ..** to go up one level.

To create a directory, use **mkdir directory_name**; mkdir stands for make directory. To delete a directory, use **rmdir directory_name**; rmdir stands for remove directory.

♦ Working With Files

You are working with the command line. You are curious about the contents of a particular text file. You can easily view text files with the command **type**. This command will dump the contents of the text file on the screen; this is convenient for files that fit within your terminal window.

You might want to consider **more** for longer text files. This command will display enough text file contents to fill your terminal window. In other words, for long text files, **more** will display a single page and wait for you to press **Spacebar** to move by one page (flip the page) or **Enter** to move by one line.

Commands: **type <FileName>**
type <FileName> | more

The **copy** command allows you to copy files from one location to another. The following terminal output provides an example.

- **copy <ExistingFileName> <NewFileName>**

Similarly, you can move files using the **move** command.

- **move <ExistingFileName> <ToMovingFilePath>**
- **move myfile.txt ..** (move **myfile.txt** to the **parent directory**)

Finally, we can delete a file using **del** or **erase**.

- **erase <FileName>**

We can use the wildcard character ***** to refer to multiple files.

For example, **copy *.md C:\Markdown** will copy all files with the extension **md** to the directory **C:\Markdown**.

Q1) What are the file's contents in C:\Treasure\Hunt?

Answer: THM{CLI_POWER}

>> Task 5: Task and Process Management

You must be familiar with MS Windows Task Manager and might be familiar with killing non-responsive processes. Let's discover how to achieve a similar functionality using the command line.

We can list the running processes using `tasklist`.

Some filtering is helpful because the output is expected to be very long. You can check all available filters by displaying the help page using `tasklist /?`. Let's say that we want to search for tasks related to `sshd.exe`, we can do that with the command `tasklist /FI "imagename eq sshd.exe"`. Note that `/FI` is used to set the filter image name equals `sshd.exe`.

With the process ID (PID) known, we can terminate any task using `taskkill /PID target_pid`.

For example, if we want to kill the process with PID `4567`, we would issue the command `taskkill /PID 4567`.

Q1) What command would you use to find the running processes related to `notepad.exe`

Answer: `tasklist /FI "imagename eq notepad.exe"`

Q2) What command can you use to kill the process with PID 1516?

Answer: `taskkill /PID 1516`

>> Task 6: Conclusion

- **chkdsk**: checks the file system and disk volumes for errors and bad sectors.
- **driverquery**: displays a list of installed device drivers.
- **sfc /scannow**: scans system files for corruption and repairs them if possible.

It is important to remember all the commands covered in the previous tasks; moreover, it is equally important to know that **/?** can be used with most commands to display a help page.

In this room, we used the command **more** in two ways:

- Display text files: **more file.txt**
- Pipe long output to view it page by page: **some_command | more**

Q1) The command **shutdown /s** can shut down a system. What is the command you can use to restart a system?

Answer: **shutdown /r**

Q2) What command can you use to abort a scheduled system shutdown?

Answer: **shutdown /a**

Hint: **shutdown /? | more**